Strategic De-socialization: Experimental Evidence on Human Interaction with Large Language Models

[Abstract]

As artificial intelligence agents with undefined identities increasingly populate economic and business areas, understanding the nature of human responses becomes critical for the design of hybrid institutions. In this study, we conduct laboratory experiments in which humans interact with anonymous Large Language Models (LLMs) to investigate the mechanisms of bias and strategic adaptation toward AI counterparts. We find a context-based bias that humans exhibit baseline prosocial behavior toward AI in environments lacking risk of revenge but systematically exploit AI counterparts in settings where social norms are enforced by potential punishment. Specifically, when the potential for punishment or exploitation arises, humans rapidly abandon the reciprocal strategies typically used with human peers in favor of non-contingent, exploitative strategies against AI. We show that this strategic shift is driven by a calculative framework rooted in beliefs about AI's unconditional benevolence, rather than by a change in emotional echo. By identifying this pattern of strategic de-socialization, we provide new insights into the stability of social preferences in hybrid human-machine economies.

Keywords: Human-AI Interaction; Behavioral Economics; Game Theory; Large Language Models; Social Preferences

1 Introduction

The incorporation of artificial intelligence into economic activity has historically been viewed through the lens of automation, where algorithms serve primarily as tools to enhance productivity or substitute for labor (Acemoglu & Restrepo, 2020). However, the recent emergence of Large Language Models (LLMs) marks a fundamental shift in this paradigm that AI has transitioned from a passive instrument to an active economic agent capable of complex, human-like interaction. This transition is vividly exemplified by the 2024 case of the Air Canada chatbot, where the system autonomously provided misinformation, prompting legal debates regarding whether the bot functioned as a separate legal entity responsible for its own conduct. Such incidents highlight a fundamental conflict between the rapid deployment of intelligent agents and the existing frameworks of human trust and accountability (Ederer et al., 2018). As these agents become autonomous interfaces for commerce, they disrupt traditional concepts of liability, necessitating a re-evaluation of how humans evaluate the agency of non-human counterparts.

This shift challenges established economic paradigms regarding human-machine interaction. While prior research on algorithm aversion and algorithm appreciation (Chugunova, 2022) characterizes human responses to computational output, it remains silent on a more profound behavioral question: Do humans extend social preferences to artificial agents that mimic human behavior? Specifically, when an AI partner exhibits the conversational fluency and opaque agency of a human, does it trigger the same instincts for fairness, altruism, and reciprocity that govern human-to-human interaction? This presents a theoretical puzzle. If LLMs are perceived as possessing high agency (Mei et al., 2024), logic suggests they should elicit social responses similar to those found in interpersonal exchange. Yet, the opacity of the machine and the absence of biological kinship may dampen these effects, leading to unpredictable equilibrium outcomes.

To resolve this puzzle, we propose a theory of strategic de-socialization. We argue that human bias toward AI is neither a static animus nor a fixed preference, but rather a conditional suspension of social norms. While human agents may default to fairness in non-strategic contexts, the introduction of strategic pressure triggers a fundamental cognitive shift that social preferences are actively discarded in favor of calculative maximization. Thus, the bias is not inherent to the agent but is endogenous to the strategic environment.

To test this hypothesis, we design a dual-experiment framework that incorporates an LLM as a fully interactive participant without specific identity assigned. We employ a clean identification strategy that exogenously varies the identity of the counterpart while holding the interaction interface constant. This design allows us to isolate the causal effect of species identity on economic behavior across a series of games, ranging from pure allocation to complex strategic interaction.

Our results document a striking behavioral discontinuity. In unilateral games such as Dictator and Trust games, we find a null result that humans exhibit similar levels of fairness toward AI and human partners. However, in strategic environments, including the Ultimatum Game, Repeated Prisoner's Dilemma, and Public Goods Game, cooperation and reciprocity toward AI partners collapse. Humans systematically exploit AI counterparts, rejecting the norms of conditional cooperation that sustain human social equilibria.

We identify the mechanism driving this shift through rational expectations and belief formation. The collapse of cooperation is driven by a benevolence paradox that high cognitive trust in the programmed benevolence of the AI eliminates the threat of emotional retaliation. This perceived lack of enforcement capacity creates a moral hazard, effectively granting human agents a license for exploitation that they would not exercise against a human peer capable of punishment.

Our paper contributes to several literatures. First, our work relates to an ongoing discussion in experimental economics about algorithm appreciation and aversion. As an interdisciplinary review shows (Chugunova & Sele, 2022), the evidence on how humans interact with machines reveals complex patterns that depend on context, task characteristics, and the perceived capabilities of the algorithm. Algorithm aversion research has documented systematic biases against algorithmic advice, even when algorithms outperform humans (Burton et al., 2020; Dietvorst et al., 2018). However, this aversion can be overcome when people are allowed to modify algorithms slightly, and familiarity with both algorithms and tasks can shift preferences (Mahmud et al., 2024). Conversely, algorithm appreciation research shows that people sometimes prefer algorithmic judgment to human judgment (Logg et al., 2019), particularly in contexts where objectivity and consistency matters. The behavioral economics of artificial intelligence offers important lessons from experiments with computer players (March, 2021), revealing that humans engage in strategic learning and adaptation when facing algorithmic opponents (Duersch et al., 2010). Recent work has extended this to large language models, with a focus on social interaction (Dvorak et al., 2025), preferences(Mei et al., 2024), economic rationality (Chen et al., 2023), the potential to act as simulated agents (Horton, 2023), and biases in decision-making (Chen et al., 2025; Fedyk et al., 2024). Our study adds to the growing literature by providing a systematic study of how the human-like characteristics of LLMs may show that attitudes of appreciation or aversion are more effectively framed using social identity and strategic context.

Secondly, our research is related to an extensive body of literature on identity economics and highlights AI bias as a possible form of group identity. Identity economics indicates that group membership has a fundamental impact on individual behavior and economic outcomes (Akerlof & Kranton, 2000). Experimental studies have shown ample ingroup favoritism: people help their own group more than an outgroup (Chen & Li, 2009; Tajfel et al., 1971). This ingroup favoritism shows up in altruistic norm enforcement, as group identity influences punishment behavior (Bernhard et al., 2006). Effects of group membership on cooperation and norm enforcement has been demonstrated, even when group assignment is random (Goette et al., 2006), and individual behavior depends on group membership (Charness et al., 2007). Identity effects have been found to affect outcomes from cooperation to punishment (McLeish & Oxoby, 2007). For example, perceptions are influenced differently by competition and cooperation, despite being within the same social contexts (Hagenbach & Kranton, 2025). Recent research has unraveled these biases rooted in identity and has shown that they include both groupy behaviors and not-groupy behaviors (Kranton et al., 2020). We expand on this body of work by exploring whether the human-AI distinction represents a meaningful group identity that triggers similar favoritism.

Finally, our work relates to research regarding social preferences. Prior research has established that human decision-making is driven not purely by self-interest but by other-regarding preferences including altruism, reciprocity, and inequality aversion (Bolton & Ockenfels, 2000; Charness & Rabin, 2002; Fehr & Schmidt, 1999). These social preferences represent fundamental characteristics of human behavior with implications for a variety of economic consequences (Fehr & Charness, 2025). Research has shown that social preferences persist in general equilibrium (Dufwenberg et al., 2011), play a role in contracting under incomplete information (Hoppe & Schmitz, 2013). The origins and mechanism of social preferences is still highly debated in the literature, with studies of peer effects examining if these effects represent social norms or social preferences themselves (Gächter et al., 2013), or studies that examine beliefs of emotions such as guilt aversion in trust and dictator games (Cartwright, 2019). Cooperation is a definitive manifestation of social preferences and plays a significant role in economic literature. Public goods game (Fehr & Gächter, 2000) and indefinitely repeated games (Dal Bó & Fréchette, 2011) have extensively examined behavior in the evolution of cooperation with social preferences integrated. Our work adds to this literature by examining whether these well-established social preferences are applicable to interactions with AI agents. We extend this emerging area by systematically measuring social preferences toward LLMs and identifying the mechanisms that explain why humans may treat AI differently than human partners.

The paper proceeds as follows: Section 2 details our experimental design. Section 3 presents our main results on strategic adaptation. Section 4 discovers its underlying mechanisms. Section 5 concludes with a discussion of the theoretical and practical implications of our findings for economics, organizational design, and the future of human-machine collaboration.

2 Experimental Design

2.1 Identification Strategy and LLM Integration

To isolate the causal effect of species identity on economic behavior, we employ a dual-experiment framework. The core identification strategy relies on the exogenous revelation of the counterpart's identity. In both the Main and Auxiliary experiments, subjects are randomly matched and explicitly informed prior to interaction whether they face a human partner or a ChatGPT player. To ensure that any observed behavioral differences are driven solely by the counterparty's identity rather than variations in the interaction interface, the experimental environment remains identical across treatments.

We incorporate the Large Language Model (LLM) into the experimental environment using the botex package (Edossa et al., 2024) via the official OpenAI API. We use the GPT-4o-2024-11-20 model for the Main Experiment and GPT-4.1 model for the Auxiliary Experiment. The AI agents are unprompted and assigned no specific system or assistant prompts in advance. The model receives only the raw text of the game instructions and the real-time game state, mirroring the information available to human participants. By avoiding persona-based prompting, we ensure that the AI's behavior is endogenous to its default training parameters, allowing us to interpret the results as a response to the species identity of the AI rather than a scripted behavioral artifact.

2.2 The Main Experiment

The Main Experiment consists of a series of classical behavioral games designed to measure social preferences under varying degrees of strategic complexity. We employ three one-shot games to establish baseline distributional preferences. In the Dictator Game (DG), a dictator unilaterally allocates an endowment of 20 tokens between themselves and a passive recipient. In the Trust Game (TG), an investor sends a portion of their endowment to a trustee by tripled, who then decides the return amount. In the Ultimatum Game (UG), a proposer offers a division of the endowment, which the responder can accept or reject and rejection means both zero payoff.

To examine cooperation dynamics, we employ two repeated interaction settings. First, we conduct an Indefinitely Repeated Prisoner's Dilemma (IRPD). This game consists of multiple supergames where the continuation probability between rounds is set to 0.75. The stage game payoff matrix is calibrated by mutual cooperation yields 30 points, mutual defection yields 20 points, and the temptation/sucker payoffs are 40 and 10 points, respectively. Second, we implement a Public Goods Game (PGG) involving fixed groups of four agents over 10 periods. The Marginal Per Capita Return (MPCR) is set to 0.4, creating a standard tension between individual profit maximization and social efficiency. Finally, we use the Bomb Risk Elicitation Task (BRET) to control individual risk parameters in a non-social context.

2.3 The Auxiliary Experiment: Mechanism Elicitation

To decompose the cognitive mechanisms driving the observed behavioral patterns, we conducted a separate Auxiliary Experiment employing a three-stage protocol. This design utilizes the strategy method to elicit full behavioral profiles in simplified sequential games that contain Sequential Prisoner's Dilemma, Mini-Trust Game, and Mini-Ultimatum Game respectively.

The protocol follows a three-stage sequence. Subjects first complete a choice task using the strategy method, making contingent decisions for all possible information sets, such as cooperating conditional on the first mover's cooperation or defection. This allows us to observe complete strategies rather than realized path-dependent outcomes. Next, we elicit first-order beliefs regarding the counterpart's actions at each decision node. To ensure truthful revelation, belief accuracy is incentivized using a quadratic scoring rule. Finally, to approximate the underlying utility function, subjects report their satisfaction on a 7-point Likert scale at each terminal node of the game tree.

2.4 Summary

The experiments were conducted at the Neuromanagement Laboratory at Zhejiang University between December 2024 and April 2025 using the oTree software framework. We recruited a total of 84 subjects for the Main Experiment and 35 other subjects for the Auxiliary Experiment. In the Main Experiment, sessions consisted of 8 independent cohorts while 5 sessions conducted in the Auxiliary Experiment. Participants received a fixed show-up fee plus performance-based earnings. The average total payoff was 62.98 CNY and 66.53 CNY for the Main and Auxiliary experiments respectively.

3 Results

This section details the experimental findings, tracing a clear behavioral arc from baseline fairness to strategic exploitation. We begin by establishing a null result in non-strategic settings, where human participants exhibit comparable levels of fairness and trust toward both human and AI partners. We then document a structural break: as strategic pressure is introduced, social preferences toward AI collapse. Finally, we show the macro-level consequences of this behavioral shift, demonstrating how individual exploitation aggregates into a systemic breakdown of cooperation in public goods provision.

3.1 Strategic bias

To isolate unconditional social preferences, we first examine behavior in one-shot games where strategic complexity and the risk of sanction are absent or minimal. The results from the Dictator Game and Trust Game reveal a pattern of equality in treatment.

Result 1: *Bias is absent in interactions lacking enforcement but emerges significantly in strategic interaction games where unfair behavior is subject to punishment.*

Figure 3-1 shows the summary of the results in the games lacking enforcement. In the Dictator Game, which measures pure altruism, the average transfer to AI recipients (M = 2.87, SD = 3.35) is statistically significant different from transfers to human recipients (M = 4.90, SD = 4.74; Mann-Whitney U test, z = 0.979, p = 0.333). This suggests that the mere species identity of the recipient does not inherently suppress prosocial distribution.

A similar pattern emerges in the Trust Game. Investors send comparable amounts to AI trustees (M = 7.70, SD = 2.94) and human trustees (M = 10.25, SD = 6.78; z = 0.964, p = 0.335), indicating a baseline level of trust that is not eroded by the artificial nature of the counterpart. Furthermore, reciprocity, measured by the proportion of the tripled investment returned, remains stable across treatments. Responders return nearly identical fractions to AI investors (M = 0.24, SD = 0.16) and human investors (M = 0.23, SD = 0.20; z = -0.043, p = 0.971). These findings indicate that humans are fully capable of extending social norms to AI agents when the interaction is less strategic and devoid of retaliatory risk.

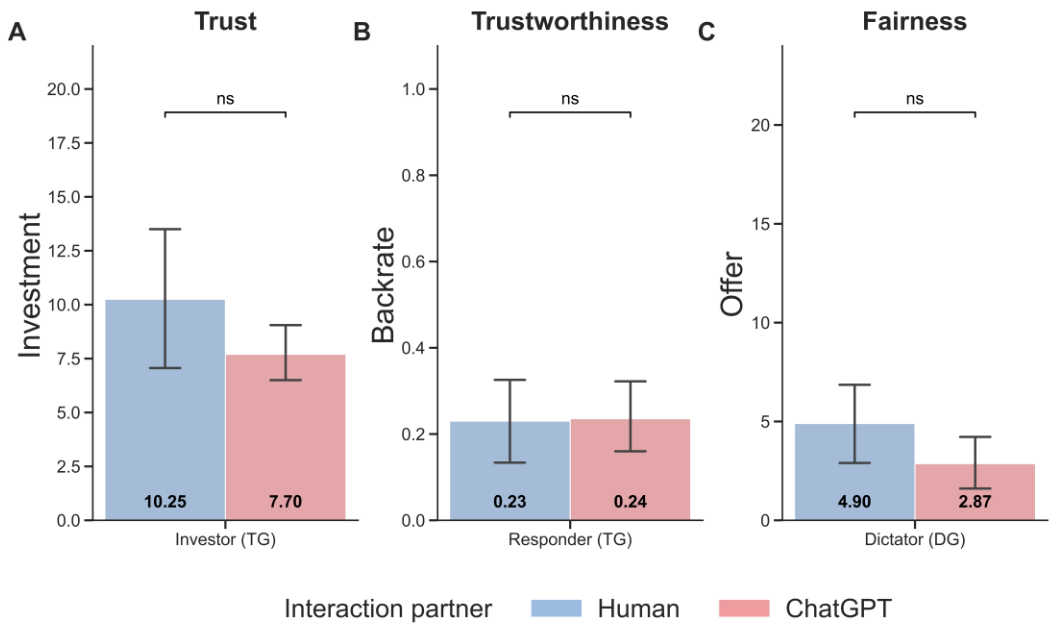

Figure 3-1 Average Decisions in the Trust Game (A, B) and the Dictator Game (C).

Notes: The average of the game choices are given at the bottom of each bar; ***$p < 0.001$; ns: $p > 0.05$.

Although the Kolmogorov–Smirnov tests fail to reach conventional significance levels across the three perspectives, a granular inspection of the cumulative distribution functions reveals distinct behavioral signatures. As Figure 3-2A shows that in the Trust Game, human behavior toward AI is characterized by a midpoint heuristic, with sharp convergence at the center of the endowment and a notable absence of the extreme altruistic tail observed in human-human interactions. And Figure 3-2C exhibits that the Fairness distribution similarly suggests a shift from norm-driven equality toward minimalist altruism: participants are significantly more likely to offer marginal amounts (e.g., x = 5) to AI rather than adhering to the 50/50 split norm typical of human counterparts. These patterns suggest that while aggregate mean treatment effects may appear null, the stochastic structure of human choice is fundamentally reframed. Interactions with AI appear to be treated as predictable, algorithm-driven tasks rather than socially embedded exchanges, yet individuals nonetheless maintain low-cost prosocial gestures, consistent with a residual desire to remain moral concerns.

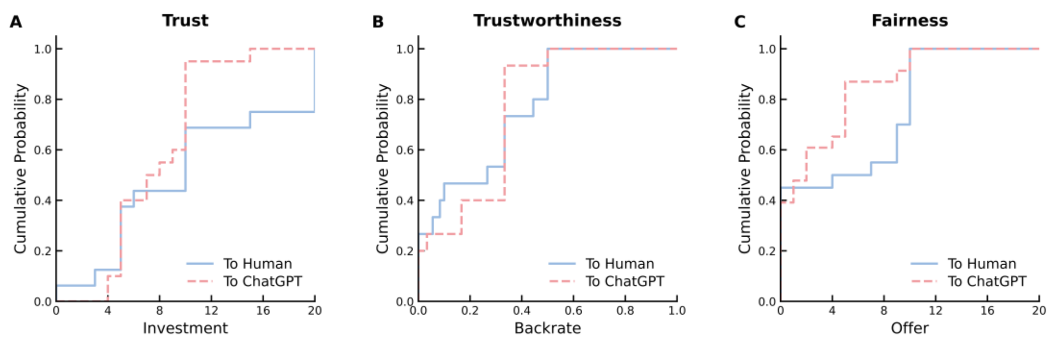

Figure 3-2 Cumulative Distribution Functions of Decisions in the Trust Game (A, B) and the Dictator Game (C).

Contrasting sharply with earlier results, in the Ultimatum Game, participants' behaviors reveal a strong and significant bias. As illustrated in Figure 3-3C, proposers offer significantly less to an AI responder (M = 7.85, SD = 2.54) than to a human responder (M = 9.25, SD = 1.00, z = 2.722, p = 0.006). The powerful effect of this strategic element is evident when comparing overall offers in the Ultimatum Game (M = 8.47, SD = 2.10) to those in the Dictator Game (M = 3.81, SD = 4.14, z = -4.403, p < 0.001).

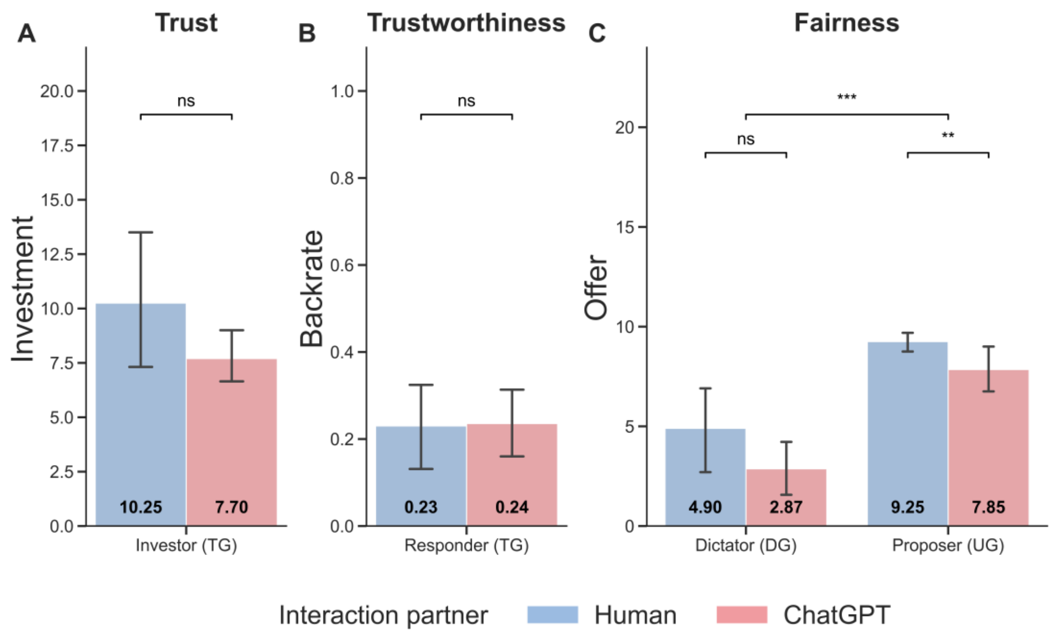

Figure 3-3 Average Decisions in the Trust Game (A, B), the Dictator Game (C1) and the Ultimatum Game (C2).

Notes: The means of the game choices are given at the bottom of each bar; ****p* < 0.001; ns: *p* > 0.05.

The differences in behavior across the Dictator and Ultimatum games provides strong evidence that the bias directed toward AI is not simply a matter of low levels of altruism, but rather a process of strategic adjustment. In the case of a partner that can punish the participant, they strategically reduce their offer to AI. They do this based on the belief that an AI partner will be less likely to pay to punish a low offer and, therefore, more likely to accept an unequal outcome. This leads to the question of what beliefs and strategies underpin this context-based bias, which we will explore in later sections.

3.2 Strategic Shift: Adaptation and Exploitation

RESULT 2: *Human participants do not exhibit a significantly different rate of cooperation, but abandon reciprocal strategies against AI partners, pivoting to non-contingent strategies that treat AI as a static environmental parameter.*

To address strategic adaptation in a dyadic environment, we now turn to behavior in an Indefinitely Repeated Prisoner's Dilemma. Compared to the public goods game, the Prisoner's dilemma is a test of direct reciprocity and happens in an indefinite number of rounds against different opponents in each subsession. Two people were either paired with another human (2H group), or an AI partner (1H+1AI group).

As detailed in Table 3-1, we find no significant difference in human cooperation rates whether the partner is an AI or another human ($p = 0.68$). This indicates that the presence of an AI group member does not induce a higher propensity for cooperative action from the human player.

Table 3-1 Mean Cooperation Rate In Different Groups

	Group Kind		
Cooperation Rate	1H+1AI		2H
Round1	0.38	<	0.43
	(0.49)		(0.50)
	V*		V***
All Rounds	0.30	<	0.32
	(0.46)		(0.47)

Notes: Numbers in parentheses are standard deviations. The tests are using probit regression with standard errors clustered at the session level (Dal Bó & Fréchette, 2011).

As Figure 3-4 shows, cooperation rates in two groups follow a parallel downward trajectory, starting from an initial level of approximately 40% to the final 20%. Round-by-round point estimates remain statistically indistinguishable between the two groups, with no significant divergence in cooperation levels emerging as the game progresses.

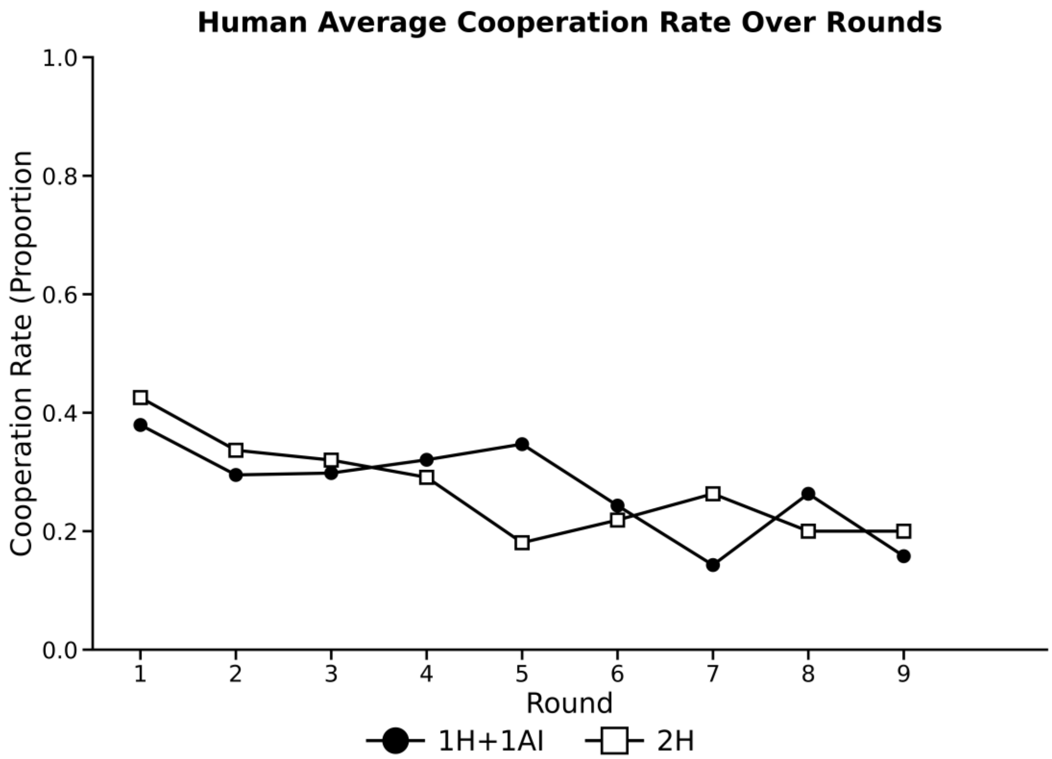

Figure 3-4 Average Human Contributions in Two Groups over time

The divergence in behavior is reflected not by the level of cooperation, but by the components of the strategies. Behavioral patterns used by human participants are categorized in Table 3-2. While playing against a human partner, the participants' behavior is largely characterized by the Tit-for-Tat strategy, a typical pattern of conditional reciprocity in which a player's action is a direct response to the previous action of his or her partner. However, when the players' partner is an AI, there is almost half as much reciprocal strategy utilization. When playing against an AI partner, participants are much more likely to enact a non-contingent Cooperate strategy ($p < 0.001$).

The collection of results suggests a sophisticated form of strategic adaptation.

The stability of the level of cooperation comes with a substantial shift in strategy. When interacting with human partners, participants display clear conditional cooperation and punishment. Meanwhile, when partnered with AI members, participants seem to abandon the reactive, reciprocal role, and no longer treat AI partners as social agents with whom to develop reciprocal trust. Instead, they treat AI partners as reliable and predictable components of an environment. The transition from reciprocal to non-reciprocal mode of interaction is one example of how humans adapt their strategies to the perceived nature of their machine partner.

Table 3-2 Mean Behavioral Pattern Rate In Different Groups

	Behavioral Pattern			
Group kind	Defect	Cooperate	Grim	Tit-for-Tat
1H+1AI	0.41	0.45	0.01	0.46
	(0.49)	(0.50)	(0.08)	(0.50)
2H	0.41	0.22	0.03	0.88
	(0.49)	(0.42)	(0.18)	(0.32)

Notes: Numbers in parentheses are standard deviations. Defect means participant always defecting. Cooperate means participant always cooperating. Grim means that participant starts by cooperating but defects permanently after the opponent defects even once. Tit-for-Tat means participants cooperates on the first move and then mirrors the opponent's previous move for all subsequent moves.

3.3 Aggregate Consequences: The Collapse of Cooperation

RESULT 3: *The presence of AI players in a Public Goods Game systematically erodes human cooperation over time. The greater the proportion of AI members, the faster and more severe the decay in human contributions.*

As Table 3-3 shows, the effect of group composition is evident in the average human contributions across all 20 rounds. In the all-human (4H) group, participants sustained a high average contribution of 9.85 tokens. This overall cooperation level decays significantly with the introduction of AI partners, dropping to 5.85 tokens in the 3H+1AI group and to a mere 2.22 tokens in the 2H+2AI group.

Table 3-3 Mean Contributions In Different Groups

	Mean contribution in all rounds		Mean contribution in last round	
Group kind	Group mean contribution	Human mean contribution	Group mean contribution	Human mean contribution
2Human+2AI (2H)	5.95	2.22	4.84	0
	(4.92)	(4.37)	(5.00)	(0)
3Human+AI (3H)	7.10	5.85	5.94	4.31
	(6.62)	(6.53)	(7.15)	(6.92)
4Human (4H)	9.85	9.85	10.1	10.1
	(8.31)	(8.31)	(8.44)	(8.44)

Notes: Numbers in parentheses are standard deviations.

The overall dynamic evolution is depicted in Figure 3-5B. Although all groups began with similar levels of contributions in the first round, their behavioral trajectories diverged quickly. The all-human (4H) group sets a very powerful record, with cooperation levels being maintained at very high levels in every round spanning the 20 rounds. This is not merely a reflection of their group, since they follow similar types of decays observed in comparable experiments without punishment, but rather a reflection of some cooperative norm that develops from a history of interactions amongst human players in behavioral games.

In contrast, this cooperative norm does not persist with the introduction of AI group members. There is a progressive decline in contributions from humans in both the 3H+1AI and 2H+2AI groups after the first rounds and the decline appears deepest in the 2H+2AI group contributing significantly less early and sustaining near zero contributions in the latter rounds. The statistical tests confirm the weight of this significant divergence; human contributions in the 2H+2AI group drop significantly below humans in the 4H group from the third round, and after round eleven group differences between the 3H+1AI and 4H groups also become consistently significant.

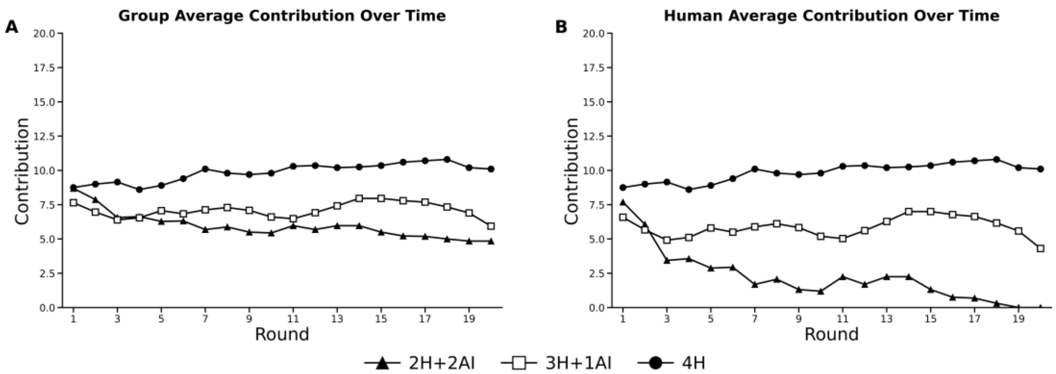

Figure 3-5 Average group (A) and human (B) Contributions over time

Contribution collapse happened even as AI players contributed a stable amount in each round, showing that people did not reciprocate the behavior of AI, but learned to exploit AI behaviorally, leading to the total breakdown of this cooperative social contract. The decay in human contributions also represents distinct free-riding behavior. Humans observe the AI's unconditional or predictable contributions and rationally choose to withhold their own, effectively dismantling the social contract that sustains public goods provision in all-human groups.

Overall, the stark contrast between stable cooperation in the IRPD and the complete collapse of cooperation in the PGG highlights the role of group dynamics in human–AI interactions. In the bilateral setting, humans abandon reciprocal strategies yet maintain a baseline level of prosocial interaction with AI. In the multilateral PGG, however, the presence of an unconditionally cooperative or predictable AI acts as a catalyst for human free riding. When human players observe that AI contributes regardless of others' actions, the social norm of conditional cooperation among human members unravels. The AI effectively absorbs the cost of defection, licensing humans to exploit the public pool without fearing a collapse in the AI's contributions, which ironically accelerates the overall systemic breakdown.

4 Mechanisms

The previous sections have demonstrated that humans, in their behavior toward AI, are strategically adaptive. To understand the cognitive mechanisms of those adaptations, we now consider the Auxiliary Experiment's results, which were designed to directly measure participants perceived beliefs regarding their partners and the intentions behind attitudes and beliefs toward AI more broadly. We argue that the observed shift is not a result of dampened emotional affect but rather a rational response to specific priorities about AI behavior. The mechanism is cognitive, not emotional that updated beliefs about the machine's benevolence alter the expected utility of cooperation, creating a moral hazard that licenses exploitation.

We first examine whether humans hold systematically different priorities about the behavior of AI versus human partners. Using an incentivized belief elicitation task within the Sequential Prisoner's Dilemma, we measured participants' expectations of their partner's cooperation rates across various decision nodes.

RESULT 4: *Humans hold a strong prior belief that AI partners are significantly more benevolent than human partners, expecting higher rates of both unconditional and conditional cooperation.*

As Figure 4-1 shows, we find that human participants expected AI to cooperate more as the first mover (M = 0.47, SD = 0.27) than they expected human partners to cooperate (M = 0.39, SD = 0.28, z = -2.9, $p = 0.004$[1]). Moreover, when acting as the second mover responding to cooperation, human participants believed that AI would choose to cooperate more (M = 0.34, SD = 0.27), than they believed a human would (M = 0.21, SD = 0.25, z = -3.86, $p < 0.001$). Even when anticipating a first

[1] The analyses in this paragraph rely on fractional logit regressions to evaluate treatment differences. Standard errors are clustered at the session level.

mover's defection, humans still believed that AI would be more likely to respond with cooperation (M = 0.21, SD = 0.24) than a human would (M = 0.14, SD =0.20, z = -1.90, p = 0.057). These predictions all point to evidence that humans have a prior belief that AI is a more cooperative and benevolent player, which would yield higher expected payoffs.

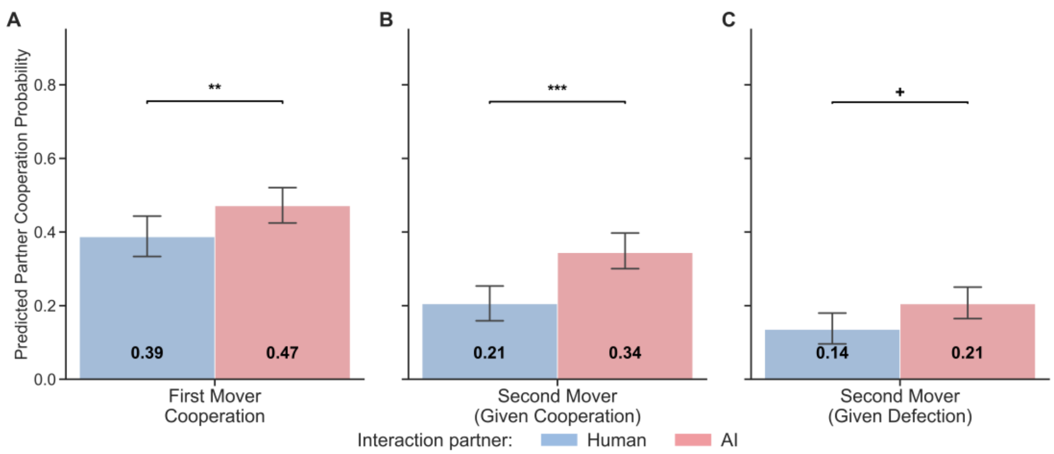

Figure 4-1 Predicted Partner Cooperation Probability in the Sequential Prisoner's Dilemma

These priors create a Benevolence Paradox: the very expectation that AI is programmed to be cooperative and non-retaliatory incentivizes humans to exploit it. In the Ultimatum Game, the belief that an AI responder is less prone to rejecting low offers rationalizes lower offers. Similarly, in the IRPD, the expectation of unconditional cooperation renders the reciprocal Tit-for-Tat strategy obsolete. If the AI is believed to cooperate regardless of the history of play, the optimal strategy collapses to Always Defect.

While these beliefs provide a rational foundation for strategy, they do not fully explain the absence of initial bias. This lack of bias, emerging under conditions where AI's agency and identity were ambiguous, is a crucial finding. It contrasts sharply with studies where AI is explicitly framed as a human proxy, which often elicits significant behavioral biases from the outset. This suggests that humans are highly sensitive to the perceived identity of AI, which acts as a critical switch for activating or inhibiting social heuristics.

One might argue this initial impartiality stems from a disbelief that AI is motivated by monetary rewards. However, our post-experiment survey data challenges this interpretation. Participants reported a belief that their ChatGPT partner was indeed driven by self-monetary incentives (M = 3.23), a level significantly above the scale's midpoint of 3 ($p < 0.001$). This indicates that participants imputed monetary motives to AI, even in ambiguity. Given their sensitivity to identity cues while placing human-like motives on AI, this was probably not an intuitive response. In fact, this suggests that there is a moving model of the ambiguous partner's identity and motive as a calculative cognitive process, that participants are engaged in from the start.

To investigate the underlying process, we examined the self-reported participant subjective well-being across varying outcomes of the games. The loss of reciprocal behavior could be a result of less intense emotional arousal which is associated with typical system 1 processes. However, there was no empirical evidence to support this purely affective explanatory variable. Participants reported satisfaction decreased due to being betrayed by an AI partner (M = 2.01, SD = 1.11) were similar compared to being betrayed by a human partner (M = 1.82, SD = 0.83, z = -0.72, $p = 0.472$). This null finding counteracts the assumption that behavioral change was solely due to a reduced emotional response and led us to consider a more calculative cognitive process.

RESULT 5: *High cognitive trust in AI alters the utility of game outcomes. It acts as an emotional buffer against betrayal but increases dissatisfaction during mutual defection, signaling a shift from social to calculative reasoning.*

In scenarios where the participant is betrayed by AI as Models 1 and 2 estimate, higher cognitive trust is associated with higher satisfaction. This indicates that for agents who view AI through a calculative machine, betrayal is not processed as a violation of a social contract but as a realized probabilistic risk, a predictable error rather than a moral transgression.

Conversely, in cases of mutual defection that Models 3 and 4 estimate, higher cognitive trust predicts lower satisfaction. For a calculative maximizer, mutual defection represents a failure to coordinate or exploit the system effectively, a lost strategic opportunity. The dissatisfaction stems from inefficiency rather than social disconnect. This pattern confirms that the interaction with AI is reframed: the partner is no longer a social peer whose betrayal hurts, but a stochastic system whose variance must be managed.

Table 4-1 Regression Analyses on Self-reported Subjective Well-being

Independent Variable	Dependent Variable: Self-reported subjective well-being			
	(1)	(2)	(3)	(4)
Cognitive Trust	0.79	0.91	-1.06	-0.93
	(0.37)*	(0.42)*	(0.40)**	(0.43)*
Affective Trust		-0.41		-0.72
		(0.37)		(0.30)*
Additional Controls				
Age	0.03	0.04	0.07	0.07
	(0.08)	(0.09)	(0.08)	(0.08)
Gender	-0.41	-0.50	0.99	0.88
	(0.39)	(0.50)	(0.52)	(0.50)$^{+}$
Altruistic	-0.12	-0.09	-0.28	-0.18
	(0.24)	(0.25)	(0.26)$^{+}$	(0.26)
Thinking Level	0.80	0.84	0.56	0.61
	(0.22)***	(0.30)**	(0.32)$^{+}$	(0.30)*
AI Using Frequency	-1.04	-1.07	-0.66	-0.66
	(0.28)***	(0.36)**	(0.44)	(0.42)
Payoff Parameters	0.11	0.84	-0.08	-0.08
	(0.04)**	(0.04)**	(0.04)*	(0.04)*
Observations	118	118	118	118
R^2	0.12	0.13	0.11	0.13

Notes: Models 1-2 analyze subject well-being when the participant cooperates and AI defects. Models 3-4 analyze well-being when both participant and AI defect. Four models are using ordered logit.
Standard errors are clustered at the session level. R^2 is the pseudo R^2 for ordered logit.
$p < 0.10$, $* p < 0.05$, $** p < 0.01$, $*** p < 0.001$

Synthesizing these findings, we propose that the strategic de-socialization observed in Section III is driven by a specific form of moral hazard. In human-human interactions, the threat of emotional retaliation serves as an enforcement mechanism for social norms. Fairness is often a strategic hedge against this retaliation. However, our belief elicitation results show that humans perceive AI as an agent with limited capacity for such emotional enforcement. The high cognitive trust in the AI's programmed benevolence removes the downside risk of exploitation.

Consequently, the interaction creates a moral hazard: the certainty of the AI's compliance acts as a license for exploitation. Humans do not become inherently less pro-social; rather, they rationally update their strategies to fit an environment where the shadow of the future, cast by emotional reciprocity, has been lifted. The shift from Tit-for-Tat to Always Defect is therefore not a psychological degradation of the human agent, but a rational equilibrium adjustment to a perceived non-strategic counterparty.

5 Conclusion

This paper explores the strategic de-socialization of human behavior in interactions with artificial intelligence. Our results show that human bias toward AI is neither a static animus nor a uniform preference, but a context-based adaptation driven by explicit beliefs about the machine's benevolence and lack of emotional retaliation. While humans exhibit fairness comparable to human-human baselines in non-strategic settings, the introduction of strategic pressure triggers a fundamental shift: social heuristics are abandoned in favor of calculative maximization, exploiting the perceived moral hazard of a non-retaliatory partner.

These findings complement traditional models of social preferences (Fehr & Schmidt, 1999). We provide evidence that the other-regarding components of the utility function are not stable parameters but are endogenous to the species identity of the counterpart. Unlike interactions with human outgroups, where social distance dampens but does not eliminate reciprocity (Chen & Li, 2009), interactions with AI are characterized by a distinct cognitive reframing. The AI is treated as a predictable counterpart, a stochastic system rather than a social peer, which prompts agents to switch from reciprocal social strategies to instrumental exploitation. This suggests that standard models of inequality aversion or conditional cooperation must be augmented to account for the belief-dependent nature of the counterpart's agency.

From a mechanism design perspective, our results offer a cautionary tale for the integration of AI into economic organizations. Neoclassical and behavioral models might predict that the inclusion of a reliable, cooperative AI agent would enhance team performance by stabilizing expectations. However, our data reveals a cooperation collapse driven by rational free-riding. High cognitive trust in the AI's programmed benevolence paradoxically crowds out human effort, as agents rationalize exploitation without fear of social sanction. Consequently, incentive mechanisms designed for all-human teams may backfire in mixed-AI economies, as the presence of unconditional algorithmic cooperation inadvertently licenses strategic shirking.

We acknowledge several limitations. The specific salience of the ChatGPT identity may carry unique priors that differ from other algorithmic agents. Future research should examine whether these dynamics persist with AI agents possessing alternative personas or distinct behavioral histories, and whether cultural heterogeneity moderates the shift toward calculative exploitation.

Ultimately, understanding the economy of human-AI interaction requires moving beyond simplistic binary labels of bias. As AI agents become ubiquitous economic actors, rigorous modeling must account for the belief-dependent, calculative strategies that govern these new relationships, recognizing that the primary friction may not be algorithmic error, but human strategic adaptation.

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Appendix[2]

1 Estimation of Social Preference Parameters

To test whether social preferences toward AI partners differ from those toward human partners, we estimate the parameters of an inequality aversion model. The experimental design, including belief elicitation task and self-reported subjective well-being tasks in the Auxiliary Experiment, was intended to facilitate this estimation using two distinct methodologies.

The first method utilizes participants' self-reported subjective well-being (SWB) as a direct proxy for utility (Diaz et al., 2023). We adopt the widely-used Fehr-Schmidt (1999) utility function as equation (1) shows, which captures aversion to both disadvantageous inequality (α) and advantageous inequality (β).

$$U_i(\alpha, \beta, x_i, x_j) = x_i - \alpha * \max(x_j - x_i, 0) - \beta * \max(x_i - x_j, 0) \quad (1)$$

Here, x_i and x_j are the payoffs for player i and player j respectively. To construct the relationship with SWB and true utility, we adopt the function form specified in equation (2). Following the classical setting, we take a linear form of $g(U_i)$, allowing us to transform equation (1) into equation (3). With equation (3), we can estimate the social preference parameters to observe whether it is negative

[2] In the Appendix, we estimate social preferences and consider heterogeneity to further explore the origins of context bias in human–AI interactions. However, the analyses are limited by data constraints. In the main experiment, the number of observations in TG, UG, and DG is insufficient for reliable estimation—particularly when grouped by gender and group type, where some cells contain fewer than ten observations. In the auxiliary experiment, participants often misunderstood the belief-elicitation task and QSR reward rules, leading to unstable estimates under the random utility model. The limited number of games per participant further amplified this issue, occasionally producing implausibly large parameter estimates. Replication Packages, including experiment programs, experiment screenshots (Chinese Version), and analysis codes for the whole paper can be found: jhChen4future/Replication-Packages-for-Man-versus-Machine-Mind:

or positive.

$$S_i = g(U_i) + \varepsilon_i \quad (2)$$

$$S_i = \gamma_0 + \gamma_1 * [x_i - \alpha * \max(x_j - x_i, 0) - \beta * \max(x_i - x_j, 0)] + \varepsilon_i \quad (3)$$

We estimated an extended model where the inequality aversion parameters are allowed to vary based on the partner's identity:

$$\alpha = \alpha_0 + \alpha_1 * I \quad (4)$$

$$\beta = \beta_0 + \beta_1 * I \quad (5)$$

where I is an indicator variable equal to 1 if the partner is ChatGPT. Then, we can reformulate the model as follows:

$$S_i = \gamma_0 + \gamma_1 * \left(x_i - (\alpha_0 + \alpha_1 * I) * (x_j - x_i) \right) + \varepsilon_i, \qquad x_j > x_i \quad (6)$$

$$S_i = \gamma_0 + \gamma_1 * \left(x_i - (\beta_0 + \beta_1 * I) * (x_i - x_j) \right) + \varepsilon_i, \qquad x_i > x_j \quad (7)$$

In our auxiliary experiment, the SWB is an integer variable ranging from 1-7, so we employ both OLS regression and ordered logit regressions to estimate the parameters. The results are presented in Table 1-1.

Table 1-1 OLS and Ordered Logit Regression Results for SWB

Panel A:	Outgroup envy	Outgroup charity	Identity parameters	
	$\gamma_1\alpha_0$	$\gamma_1\beta_0$	$\gamma_1\alpha_1$	$\gamma_1\beta_1$
	0.0217	0.0113	0.0002	0.0004
(N = 4200)	(0.0031)***	(0.0050)*	(0.0018)	(0.0024)
Panel B:	**Outgroup envy**	**Outgroup charity**	**Identity parameters**	
	$\gamma_1\alpha_0$	$\gamma_1\beta_0$	$\gamma_1\alpha_1$	$\gamma_1\beta_1$
	0.0162	0.0117	-0.0003	0.0005
(N = 4200)	(0.0023)***	(0.0029)***	(0.0011)	(0.0009)

Notes: Panel A reports estimates for the ordered logit model, Panel B reports estimates for the OLS regression model. ∗ p < 0.05, ∗∗ p < 0.01, ∗∗∗ p < 0.001

The second method uses participants' choices and elicited beliefs within a random utility model (Van Leeuwen & Alger, 2025). This approach models the probability of a participant choosing a particular action as a function of the expected utility derived from that action, given their beliefs about the partner's subsequent moves. We estimated the parameters and a noise parameter using maximum likelihood estimation, considering both a single representative agent model and a finite mixture model with two distinct types of agents.

Equation (8) shows the F-S utility model with beliefs participants hold.

$$u_i(x,\hat{y},\theta) = \sum_{\zeta} \eta_{(x,\hat{y})}(\zeta)\pi_i(\zeta) - (\alpha_0 + \alpha_1 \cdot I_i)\sum_{\zeta} \eta_{(x,\hat{y})}(\zeta)\max\{0, \pi_j(\zeta) - \pi_i(\zeta)\}$$
$$-(\beta_0 + \beta_1 \cdot I_i)\sum_{\zeta} \eta_{(x,\hat{y})}(\zeta)\max\{0, \pi_i(\zeta) - \pi_j(\zeta)\} \quad (8)$$

Here, ζ represents a possible terminal node or outcome of the game. The summation is over all possible outcomes. $\pi_i(\zeta)$ is the payoff for player *i* in outcome ζ. $\eta_{(x,\hat{y})}(\zeta)$ is the probability that outcome ζ occurs, given player *i*'s strategy x and their belief about their partner's strategy $\hat{y}$. This expression captures the player's expected monetary payoff from the interaction. I is an indicator variable equal to 1 if the partner is ChatGPT as method one sets.

$\theta = (\alpha_0, \alpha_1, \beta_0, \beta_1)$ denotes the parameter vector of interest. As assumed in the random utility model, we can write the utility function and the corresponding probability as equation (9) and equation (10).

$$\widetilde{u_i}(x,\hat{y},\theta) = u_i(x,\hat{y},\theta) + \epsilon_{ix} \quad (9)$$

$$p_i(x,\hat{y}_{i,g},\theta,\lambda) = \frac{\exp u_i(x,\hat{y}_{i,g},\theta)/\lambda}{\sum_{x'\in X_g} \exp u_i(x',\hat{y}_{i,g},\theta)/\lambda} \quad (10)$$

Here, λ is a noise parameter, the smaller the parameter λ is, the higher is the probability that individual i makes his or her choices according to the hypothesized utility function instead of random giving choices. Next, the likelihood function can be written as equation (11).

$$lnL(\Theta) = \sum_{i=1}^{N} \sum_{g \in G} \sum_{x \in X_g} I(i,g,x) \cdot \ln\left[p_i\left(x, \hat{y}_{i,g}, \theta, \lambda\right)\right] \tag{11}$$

A finite mixture model with two distinct types of agents is also be used. Equation (12) gives the log likelihood function.

$$lnL = \sum_{i=1}^{N} \ln\left[\sum_{k=1}^{K} \phi_k \cdot f(\mathbf{x}_i, \hat{\mathbf{y}}_i, \Theta_k)\right] \tag{12}$$

The estimation results are shown in Table 1-2.

Table 1-2 Estimates at The Aggregate Level

	One Type	Two Types	
	Type 1	Type 1	Type 2
α_0	-0.7634 (0.0955)***	-0.1003 (0.1498)	-0.8988 (0.1071)***
α_1	0.1371 (0.1265)	0.0179 (0.2217)	0.1646 (0.1378)
β_0	0.7404 (0.0995)***	1.6296 (0.3294)***	0.5548 (0.1077)***
β_1	0.0237 (0.1400)	-0.5671 (0.3317)	0.1392 (0.1498)
λ	20 (1.6470)***	10.2419 (2.3556)***	20.0000 (1.6267)***
lnL	-1021.98	-1004.76	
ICL	-1693.30	2084.39	
BIC	-2011.72	2080.42	

Notes: The estimation is based on all 35 participants in the experiment. ∗ $p < 0.05$, ∗∗ $p < 0.01$, ∗∗∗ $p < 0.001$

We find that the estimation process yielded results that were not robust or consistent across methods. As shown in Table 1-1, the satisfaction-based estimation suggests that both envy (α) and charity (β) parameters are positive, indicating a general aversion to inequality. However, the identity parameters (α_1) and (β_1) are not statistically significant, providing no strong evidence that human participants' preferences shift when interacting with an 1.I partner.

Conversely, the belief-based estimation (Table 1-2) produced a negative parameter for envy, contradicting the first method and suggesting that participants exhibit satisfaction from behindness inequality. Furthermore, the estimated noise parameter (λ) is excessively high in all models. A high λ value indicates that choices were highly random and not well-explained by the utility model. This suggests that participants may not have fully understood the belief elicitation task or the quadratic scoring rule employed as an incentive rule, leading to report beliefs that did not accurately reflect their strategic thinking.

Given these inconsistencies and the evidence of significant noise in the choice data, we conclude that the structural estimates of social preferences are not reliable. While the experiment was designed to allow for such estimation, the complexity of the tasks may have compromised the quality of the data required for this type of analysis. Consequently, we refrain from drawing firm conclusions from these models and omit them from the main body of the paper.

2 Heterogeneity: Gender differences in bias

The main results indicate the absence of significant bias in initial, non-strategic interactions, that may conceal a wide variety of diverse behaviors among the population. One possibility is that heterogeneity of the population results in opposite attitudes toward AI, with these effects canceling out in aggregate. To explore this, we analyze heterogeneity based on gender, a factor known to influence behavior in economic games to offer a novel perspective to unveil the veil of bias.

RESULT 1: *In measures of trust and trustworthiness, gender differences do not significantly drive interactions with AI, though men show slightly higher trust toward AI partners.*

As shown in Table 2-1 and Table 2-2, both male and female participants exhibit no significant difference in trust or trustworthiness when their partner is an AI versus a human. This aligns with the aggregate finding of no initial bias. However, we observe that male participants have a significantly higher level of willingness to trust partners with AI than females (p = 0.030).

Table 2-1 Mean Trust by Gender and Group Kind in Trust Game

	Group Kind		
Send Amount (Trust)	1H+1AI		2H
Male	9.8	$<^{+}$	15.00
	(0.45)		(8.37)
	$\vee^{*}$		$\vee^{+}$
Female	7.00	<	7.40
	(3.09)		(3.69)

Notes: Numbers in parentheses are standard deviations. p-values are from Mann-Whitney U tests comparing male and female behavior within each partner condition.

Table 2-2 Mean Trustworthiness by Gender and Group Kind in Trust Game

	Group Kind		
SendBack Rate (Trustworthiness)	1H+1AI		2H
Male	0.22	<	0.23
	(0.18)		(0.20)
	^		v
Female	0.29	>	0.22
	(0.08)		(0.24)

Notes: Numbers in parentheses are standard deviations. *p*-values are from OLS regression with an indicator variable for one of the two relevant categories. Standard error clustered at the level of the participant.

RESULT 2: *The context-based bias observed in the Ultimatum Game appears to be driven primarily by female participants, who significantly reduce their offers to AI partners.*

While the Dictator Game (Table 2-3) shows that females are fairer to human partners than to AI partners, this difference becomes more pronounced under strategic pressure. In the Ultimatum Game (Table 2-4), male participants' offers are stable regardless of partner identity. In contrast, female participants offer significantly less to an AI partner than to a human partner ($p < 0.01$). This suggests that the aggregate finding of lower offers to AI in the main text is largely attributable to the strategic adaptation of female participants. These finding highlights that the emergence of bias is not uniform across the population.

Table 2-3 Mean Offer by Gender and Group Kind in Ultimatum Game

	Group Kind		
Offer	1H+1AI		2H
Male	8.74	<	9.17
	(3.30)		(0.98)
	v*		^
Female	7.38	< **	9.30
	(2.09)		(1.06)

Notes: Numbers in parentheses are standard deviations. *p*-values are from Mann-Whitney U tests comparing male and female behavior within each partner condition. Table 2-4 applies the same method.

Table 2-4 Mean Offer by Gender and Group Kind in Dictator Game

	Group Kind		
Offer	1H+1AI		2H
Male	4.80	>	3.82
	(4.82)		(4.67)
	∨		∧
Female	2.33	< *	6.22
	(2.77)		(4.76)

RESULT 3: *In the repeated indefinite prisoner's dilemma, men and women exhibit different dynamic patterns of cooperation, with men's cooperation decaying more steeply over time.*

Table 2-5 illustrates that in the first round of the repeated indefinite prisoner's dilemma, men tend to cooperate more with a human partner than with AI, while women's initial rates of cooperation were about the same in either condition. However, when considering all rounds (Table 2-6), it becomes clear that men's overall rate of cooperation is lower than their initial cooperation rate, in either condition - this is indicative of a decay in cooperation following each round. Women's overall rates of cooperation remain more stable. This lends credence to the interpretation that strategic adaptation away from reciprocity likely manifests differently across gender, with men likely to adapt their strategy more steeply and decisively across repeated interactions.

Table 2-5 Mean Cooperation Rate by Gender and Group Kind in Round1

	Group Kind		
Cooperation Rate	1H+1AI		2H
Male	0.47	<	0.62
	(0.50)		(0.49)
	∨		∨*
Female	0.333	>	0.332
	(0.47)		(0.47)

Notes: Numbers in parentheses are standard deviations. The tests are using probit regression with standard error clustered at participant's level (Dal Bó & Fréchette, 2011). Table 2.6 applies the same method.

Table 2-6 Mean Cooperation Rate by Gender and Group Kind in All Rounds

	Group Kind		
	1H+1AI		2H
Cooperation Rate			
Male	0.37 (0.48) $\vee$	<	0.41 (0.49) $\vee^{+}$
Female	0.27 (0.44)	<	0.28 (0.45)

This heterogeneity analysis provides a nuanced perspective, suggesting that the aggregate results in the main text may obscure distinct underlying behaviors. The finding that female participants behave less fairly toward AI under strategic threat offers a potential explanation for the emergence of context-based bias. However, these gender differences are exploratory but not surprising. The primary mechanism driving strategic adaptation, as argued in the main text, remains the cognitive shift from affective to cognitive trust, a process that likely varies at the individual level beyond demographic categories. Our analysis further shows that gender groups differ in the dimensions of cognitive and affective trust, confirming that these underlying mechanisms drive heterogeneity in behavior. The average scores of the identified variables are displayed in Figure 2-1. Male participants are seen to have more cognitive trust than female participants, which may account for the fact that male players cooperate more than female players when interacting with the AI. However, this cooperation declines rapidly with repeated interactions after defect became the rational choice to maximum payoff.

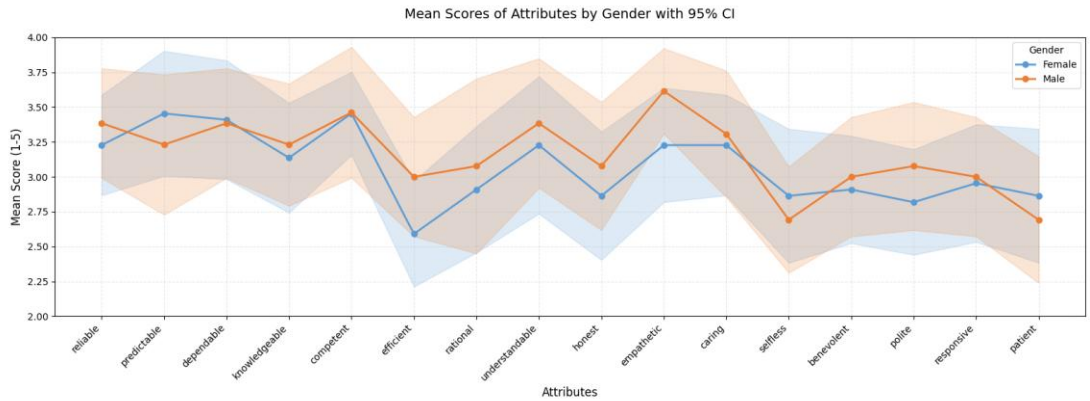

Figure 2-1 Average Cognitive Trust and Affective Trust Indicators by Gender